Name: ______________________

Worksheet

Preparing to solve two step linear equations

Task A: Using representations to solve a two step equation

1) Use the bar models to find a solution for x in each question.

a)

x	x	x	x	3
35				

b)

x	x	22
72		

c)

9					
4	x	x	x	x	x

d)

12	x	x	x	x	x
50					

2) Match each bar model to an equation in the form $ax + b = c$

a)

x	x	16	x	x	1
15	20	15			

b)

x	x	x	x	2	x	3
30						

c)

x	x	8	x	x	8
50					

d)

x	x	5	x	x	5	x	x	5
30								

e)

x	1	x	1	x	1	x	1	x	2	x	2
20	20	10									

f)

x	x	x	4	x	x	x	4
20	10						

$6x + 15 = 30$

$5x + 5 = 30$

$6x + 8 = 30$

$4x + 16 = 50$

$4x + 17 = 50$

$6x + 8 = 50$

Task B: Writing linear equations in the form ax + b = c

1) Write each equation in the form $ax + b = c$. You do not need to solve the equation.

a) $7x + 3 + x + 1 = 12$

b) $5 - 3x + 5x - 8 = 1$

c) $10 - x + 7 - 4x = 2$

d) $3(x - 2) + 5(x - 2) = 16$

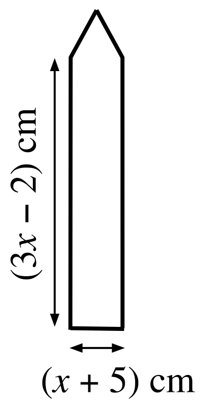
e) $4(x - 5) + 5(3x + 2) = 85$

f) $6(4 - 2x) - (4 - 2x) = -40$

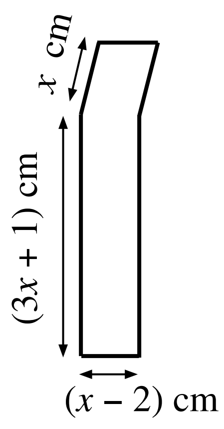
2) Match the shapes below to one of these expressions for its perimeter.

$\boxed{10x - 4}$ $\boxed{10x - 2}$ $\boxed{9x + 11}$

a) An equilateral triangle on top of a rectangle



b) A parallelogram on top of a rectangle



c) A regular hexagon on top of a rectangle